A proper spectral modification violating Efimov rigid-fibre invariance

Theorem. *Let*

$$R := \mathbf{Q}[[t]], \qquad J := \bigoplus_{n\geq 1} \Sigma^n(R/t^n), \qquad A := R \oplus J,$$

where A is the split square-zero E_∞-extension, and put $K := R[1/t] \simeq A[1/t]$. Then $\operatorname{Spec} R \to \operatorname{Spec} A$ *is proper and locally almost of finite presentation and is an equivalence over $D(t)$, but*

$$\operatorname{Rig}(\operatorname{Mod}_A^{t\text{-nilp}}) \otimes_{\operatorname{Mod}_A} \operatorname{Mod}_K \longrightarrow \operatorname{Rig}(\operatorname{Mod}_R^{t\text{-nilp}}) \otimes_{\operatorname{Mod}_R} \operatorname{Mod}_K \tag{1}$$

is not fully faithful.

Proof. Every bounded truncation of J is a finite sum of perfect R-modules, hence $J \in \operatorname{APerf}(R)$; therefore $A \to R$ is locally almost of finite presentation by [2, Lem. 2.20(2)]. Its classical truncation is the identity, hence the induced spectral morphism is proper, and $J[1/t] \simeq 0$ gives $A[1/t] \simeq R[1/t]$.

Let

$$\Phi : \operatorname{Mod}_A^{t\text{-nilp}} \longrightarrow \operatorname{Mod}_R^{t\text{-nilp}}$$

be scalar extension, and let Res be its right adjoint. This induces an adjunction $F \dashv G$ on rigidifications. Let a_A, a_R be the standard maps from rigidification and Λ_A, Λ_R their fully faithful symmetric-monoidal right adjoints. Functoriality and right mates give

$$a_R F \simeq \Phi a_A, \qquad G\Lambda_R \simeq \Lambda_A \operatorname{Res}. \tag{2}$$

The usual unit-and-mate calculation identifies the unit at the tensor unit with $\Lambda_A(A \to R)$; hence

$$\operatorname{fib}\big(\mathbf{1} \to GF(\mathbf{1})\big) \simeq \Lambda_A(J). \tag{3}$$

After scalar extension to K, the fibre is

$$L_t\Lambda_A(J) := \operatorname{colim}\big(\Lambda_A(J) \xrightarrow{t} \Lambda_A(J) \xrightarrow{t} \cdots\big). \tag{4}$$

Set $Q := \bigoplus_{m\geq 1} \Sigma^{-m} R$, with A-action through $A \to R$, and put $\mathcal{D} := (\operatorname{Mod}_A^{t\text{-nilp}})^{\omega_1}$. The shifts make J, Q derived t-complete and ω_1-compact. Efimov's predual description gives

$$\operatorname{Map}\big(Q^{op}, \Theta_A L_t\Lambda_A(J)\big) \simeq \operatorname{Map}_{\operatorname{Mod}_A^{t\text{-nilp}}}\big(\mathbf{1}, J\widehat{\otimes}_A Q\big)[1/t], \tag{5}$$

where $\Theta_A : \operatorname{Rig}(\operatorname{Mod}_A^{t\text{-nilp}}) \hookrightarrow \operatorname{Ind}(\mathcal{D}^{op})$ is the canonical fully faithful functor; this is the predual description of [1, Thm. 4.2(ii)–(iii) and Prop. 1.34], with colimit compatibility from [1, Prop. 1.27].

The R-module R is a retract of $R \otimes_A R$ in Mod_R. Hence $J \otimes_R Q$ is a retract of $J \otimes_A Q$. Its equal-index summands give a further retract

$$D := \bigoplus_{n\geq 1} R/t^n \;\longrightarrow\; J \otimes_A Q. \tag{6}$$

Thus $\Lambda_t D$ is a retract of $J\widehat{\otimes}_A Q$. A direct computation gives

$$(\Lambda_t D)[1/t] \not\simeq 0. \tag{7}$$

Consequently the right side of (5) is nonzero, so $L_t\Lambda_A(J) \not\simeq 0$. Therefore the localized adjunction unit is not invertible, and (1) is not fully faithful. ☐

References

[1] A. I. Efimov, *Localizing invariants of inverse limits*, arXiv:2502.04123v2, Props. 1.27, 1.34, Thm. 4.2 and Cor. 4.4.

[2] L. Brantner, A. Magidson, and J. Nuiten, *Formal Integration of Derived Foliations*, arXiv:2502.05257, Lem. 2.20(2).